1. Bias Identification — Awareness Stage

User Story
As a student developing an academic argument, I want my reasoning to be challenged for hidden bias, so that I can recognize assumptions I didn't notice.

Acceptance Criteria

- Given I submit written reasoning, when it is evaluated, then at least one potential bias or assumption is identified
- Given bias is identified, then it is clearly tied to a specific claim
- Given feedback is provided, then I can acknowledge whether the bias is valid
- Given I accept the bias, then it is documented within the session

Supports: KR1 – Bias & Assumption Recognition

2. Logical Fallacy Detection — Awareness

User Story
As a student refining an argument, I want logical fallacies identified, so that I can strengthen my reasoning.

Acceptance Criteria

- Given an argument draft, when reviewed, then logical fallacies are flagged
- Given a fallacy is flagged, then it includes an explanation
- Given the explanation, then I can determine whether revision is needed
- Given revision occurs, then it is tracked as reasoning-based change

Supports: KR1, KR2

3. Counterargument Generation — Awareness

User Story
As a student unsure of opposing views, I want credible counterarguments presented, so that I understand how my claims could be challenged.

Acceptance Criteria

- Given a claim, when evaluated, then at least one opposing argument is produced
- Given opposition is presented, then it reflects credible reasoning
- Given the counterargument, then I can assess its strength
- Given I respond, then my rebuttal is captured

Supports: KR2, KR4

4. Reasoning Depth Evaluation — Awareness Stage

User Story
As a student under time pressure, I want my reasoning depth evaluated, so that I know whether my argument is surface-level.

Acceptance Criteria

- Given I submit an argument draft, when it is evaluated, then reasoning depth is assessed
- Given shallow reasoning is detected, then gaps are identified
- Given gaps are identified, then they are clearly tied to specific claims
- Given feedback is provided, then I can revise those areas

Supports: KR2 – Reasoning-Based Argument Revision

5. Evidence Strength Testing — Awareness

User Story
As a student defending a claim, I want my evidence stress-tested, so that I know if it is sufficient.

Acceptance Criteria

- Given supporting evidence, when evaluated, then strength is assessed
- Given weak evidence, then insufficiencies are explained
- Given insufficiencies, then stronger opposition is presented
- Given review, then revisions are measurable

Supports: KR2, KR3

6. Assumption Surfacing — Awareness

User Story
As a student forming conclusions, I want unstated assumptions surfaced, so that I can validate or revise them.

Acceptance Criteria

- Given reasoning, when analyzed, then implicit assumptions are identified
- Given assumptions are listed, then each links to a conclusion
- Given I review them, then I can confirm or reject each
- Given confirmation, then revisions are tracked

Supports: KR1, KR2

7. Argument Weakness Identification — Awareness Stage

User Story
As a student revising before submission, I want the weakest parts of my argument identified, so that I know where revision effort matters most.

Acceptance Criteria
• Given I submit a full argument, when it is reviewed, then weak claims are flagged
• Given weaknesses are flagged, then each includes reasoning risk explanation
• Given risks are identified, then I can prioritize revisions
• Given revisions occur, then changes are recorded

Supports: KR2 – Reasoning-Based Argument Revision

8. Opposition Simulation — Decision

User Story
As a student deciding whether my argument is ready, I want simulated opposition, so that I can test defensibility.

Acceptance Criteria

- Given an argument, when challenged, then opposition reflects credible viewpoints
- Given opposition, then it targets core claims

- Given I respond, then rebuttals are evaluated
- Given evaluation, then readiness confidence is measurable

Supports: KR4

9. Rebuttal Strength Evaluation — Decision

User Story
As a student preparing to defend my work, I want my rebuttals evaluated, so that I know if they withstand criticism.

Acceptance Criteria

- Given rebuttals, when analyzed, then their strength is assessed
- Given weak rebuttals, then improvement areas are identified
- Given improvements, then I can revise
- Given revision, then change is tracked

Supports: KR2, KR4

10. Reasoning Gap Identification — Awareness

User Story
As a student organizing my thoughts, I want reasoning gaps identified, so that my argument flows logically.

Acceptance Criteria

- Given structured reasoning, when evaluated, then logical gaps are identified
- Given gaps, then missing links are described
- Given descriptions, then I can address them
- Given changes, then revisions are documented

Supports: KR2

11. Defensibility Confidence Measurement — Decision Stage

User Story
As a student nearing submission, I want my argument's defensibility assessed, so that I understand how prepared it is for opposition.

Acceptance Criteria
• Given my argument has been challenged, when assessment occurs, then defensibility confidence is measured
• Given confidence is measured, then it reflects opposition outcomes
• Given low confidence is detected, then risk areas are identified
• Given revisions occur, then confidence can be reassessed

Supports: KR4 – Opposition Readiness Confidence

12A. Revision Detection — Action Stage

User Story
As a student revising my argument, I want substantive reasoning revisions detected, so that I know where my argument changed.

Acceptance Criteria
• Given I revise my argument, when evaluated, then substantive changes are identified
• Given changes are identified, then they are linked to challenged claims
• Given revisions are minor, then they are distinguished from substantive ones
• Given changes are confirmed, then they are logged

Supports: KR2 – Reasoning-Based Argument Revision

12B. Rigor Improvement Assessment — Outcome Stage

User Story
As a student revising my work, I want my argument's rigor re-evaluated after revision, so that I know if changes strengthened my reasoning.

Acceptance Criteria
• Given revisions are completed, when reassessed, then rigor is evaluated
• Given reassessment occurs, then it compares pre- and post-revision states
• Given improvement exists, then it is indicated
• Given results are generated, then they are stored

Supports: KR3 – Perceived Rigor & Defensibility Improvement

13. Bias Reduction Reflection — Outcome

User Story
As a student after revision, I want to reflect on biases I corrected, so that I improve future reasoning.

Acceptance Criteria

- Given session completion, then identified biases are summarized
- Given summary, then it shows what changed
- Given changes, then learning impact is captured
- Given capture, then it contributes to improvement metrics

Supports: KR1

14. Rigor Improvement Self-Assessment — Outcome Stage

User Story
As a student completing a session, I want to rate how much my argument improved, so that I can reflect on its rigor.

Acceptance Criteria
• Given a session ends, when prompted, then I can rate rigor improvement
• Given I submit a rating, then it reflects perceived reasoning improvement
• Given ratings are collected, then they are stored
• Given stored ratings, then they contribute to rigor perception metrics

Supports: KR3 – Perceived Rigor & Defensibility Improvement

15. Opposition Effectiveness Rating — Outcome Stage

User Story
As a student engaging with opposition, I want to rate how effectively it challenged me, so that I can evaluate its intellectual value.

Acceptance Criteria
• Given opposition is experienced, when prompted, then I can rate its effectiveness
• Given ratings are submitted, then they reflect thinking depth impact
• Given feedback is collected, then it is stored
• Given stored feedback, then it contributes to satisfaction measurement

Supports: Supporting KPI – Post-Session Rating

16. Submission Risk Identification — Decision Stage

User Story
As a student before submission, I want unresolved reasoning risks identified, so that I avoid submitting weak arguments.

Acceptance Criteria
• Given my argument is reviewed, when risks exist, then they are listed
• Given risks are listed, then they are tied to specific claims
• Given risks are identified, then I can revise
• Given revisions occur, then risks update

Supports: KR4 – Opposition Readiness Confidence

17. Multi-Session Reasoning Progression — Outcome Stage

User Story
As a student using the tool repeatedly, I want my reasoning improvements summarized across sessions, so that I recognize growth in rigor.

Acceptance Criteria
• Given multiple sessions occur, when analyzed, then improvements are aggregated
• Given aggregation occurs, then bias reductions are tracked
• Given tracking exists, then rigor trends are visible
• Given trends are visible, then progress is measurable

Supports: KR1 – Bias Recognition
Supports: KR3 – Rigor Improvement

18. Intellectual Pushback Exposure — Awareness

User Story
As a student used to agreeable feedback, I want exposure to structured pushback, so that my ideas are genuinely tested.

Acceptance Criteria

- Given argument submission, then pushback is provided
- Given pushback, then it challenges core logic
- Given challenge, then I must respond or revise
- Given response, then impact on reasoning is measurable

Supports: KR2, KR4